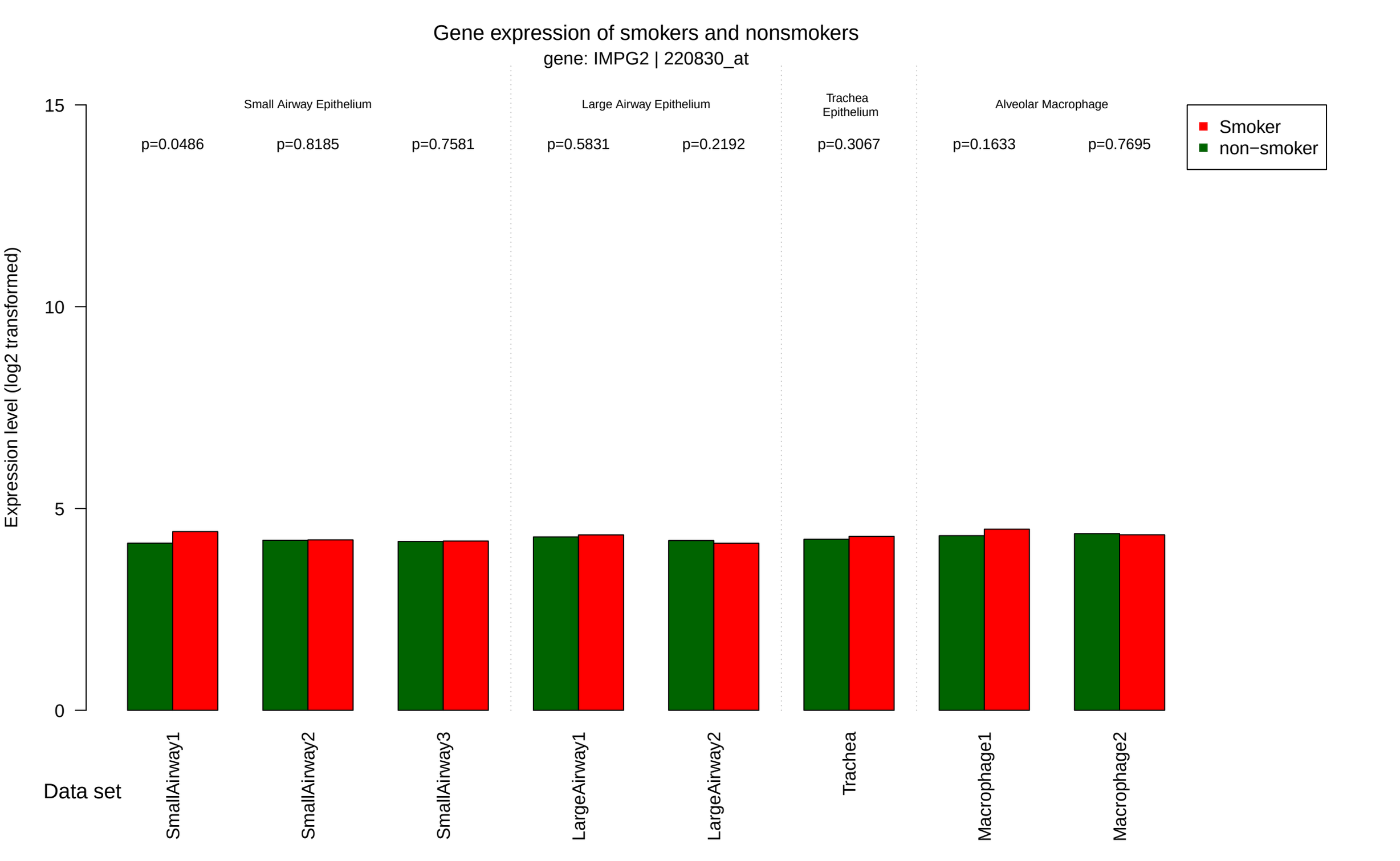
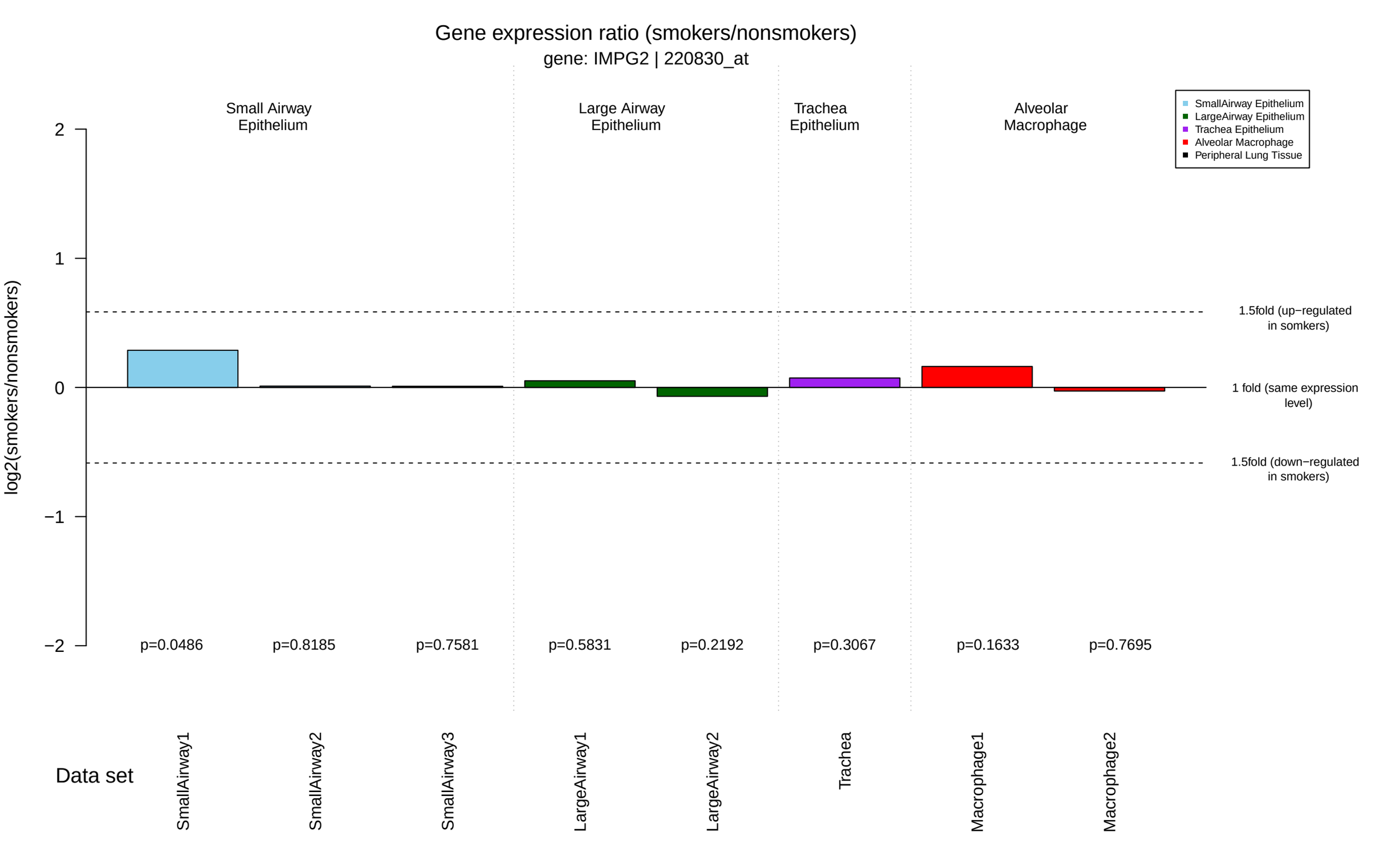
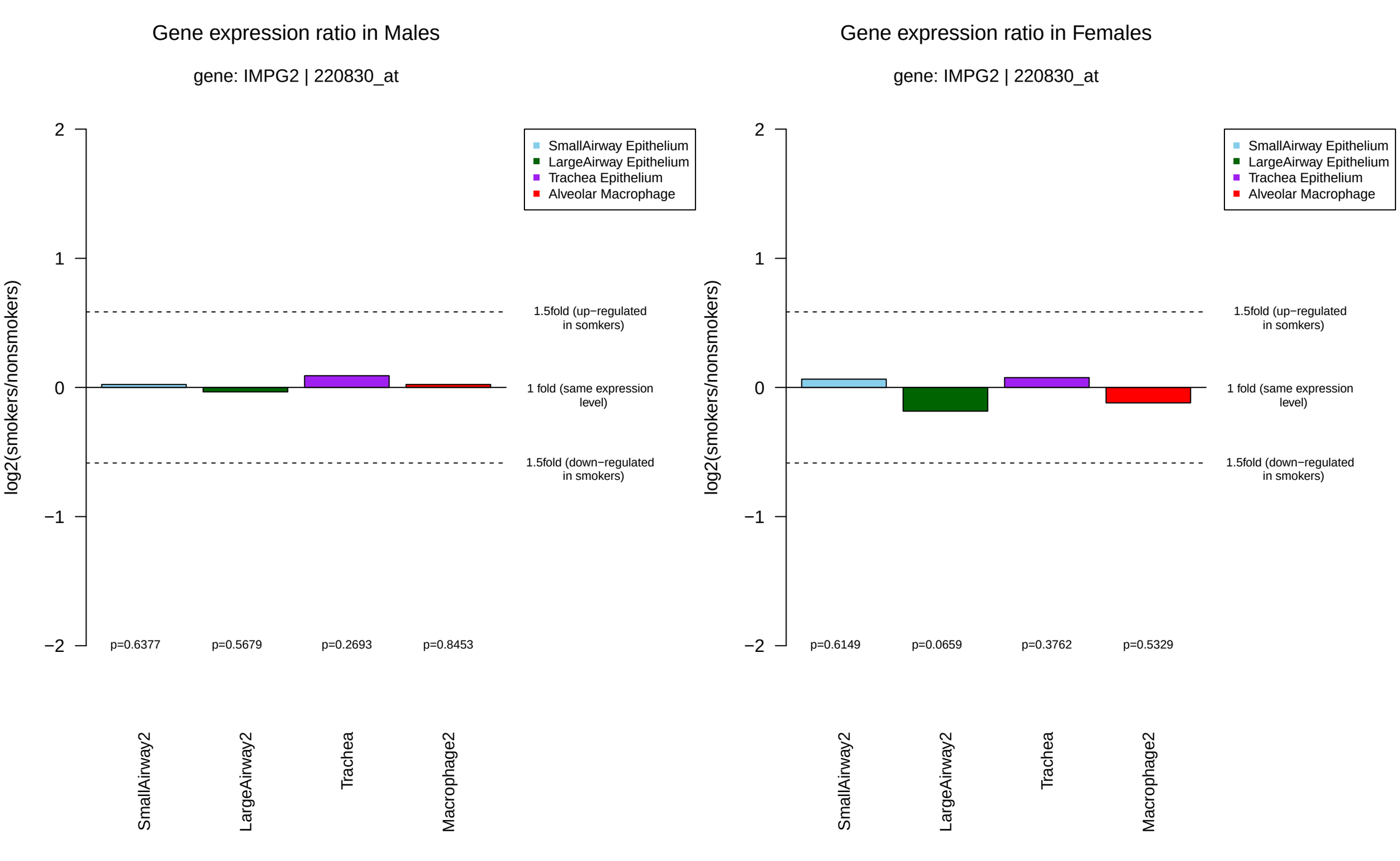
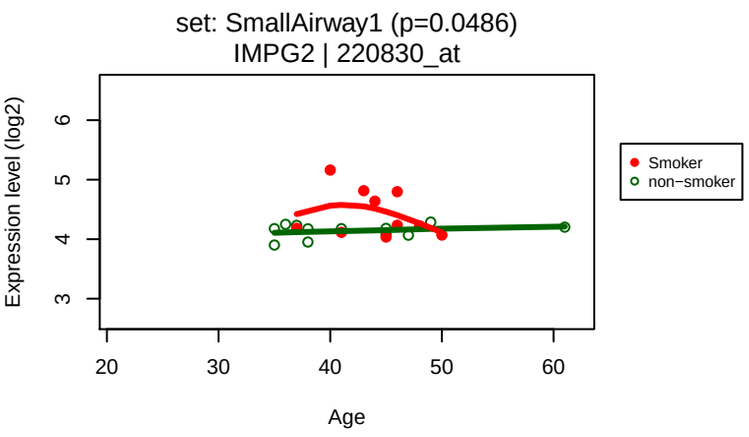
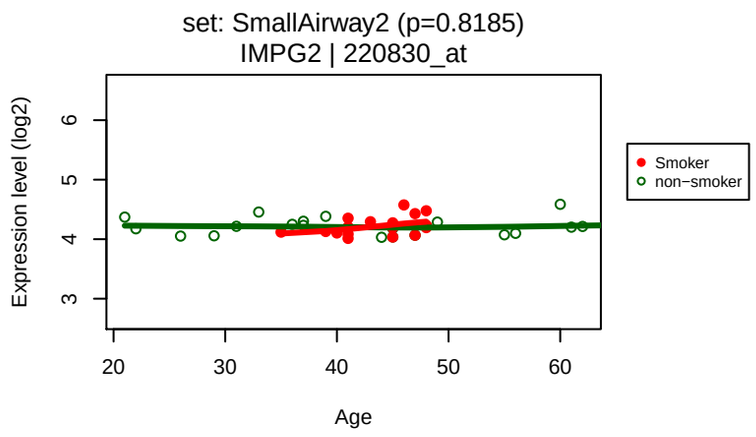
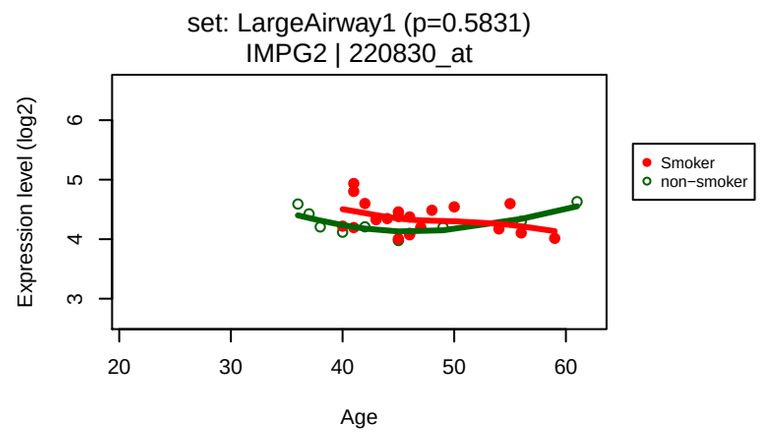

Gene expression of smokers and nonsmokers
gene: IMPG2 | 220830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0486
p=0.8185
p=0.7581
p=0.5831
p=0.2192
p=0.3067
p=0.1633
p=0.7695
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

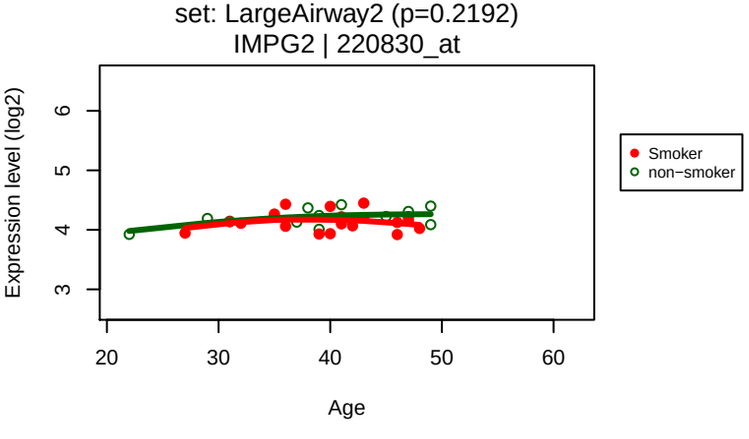

Gene expression ratio (smokers/nonsmokers)
gene: IMPG2 | 220830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0486
p=0.8185
p=0.7581
p=0.5831
p=0.2192
p=0.3067
p=0.1633
p=0.7695
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

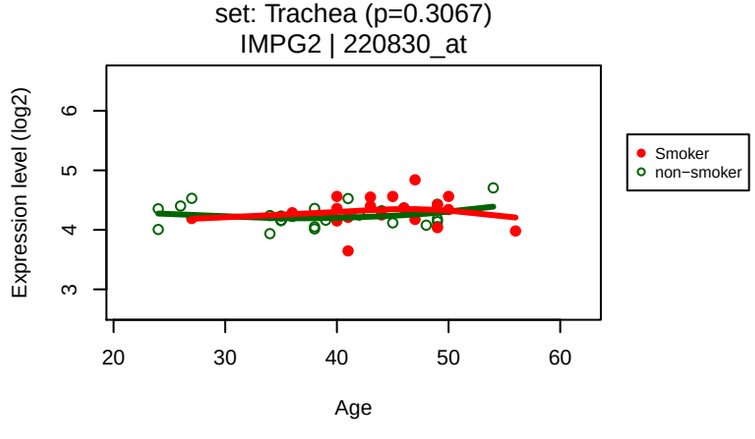

Gene expression ratio in Males
gene: IMPG2 | 220830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6377
p=0.5679
p=0.2693
p=0.8453
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IMPG2 | 220830_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6149
p=0.0659
p=0.3762
p=0.5329
SmallAirway2
LargeAirway2
Trachea
Macrophage2

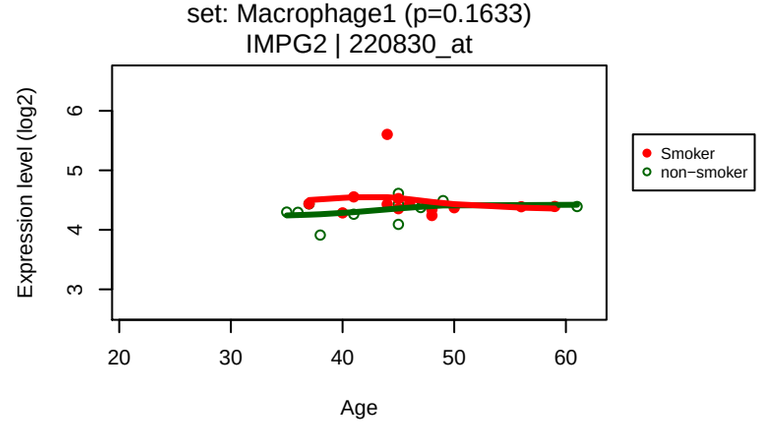

set: SmallAirway1 (p=0.0486)
IMPG2 | 220830_at
Expression level (log2)
Age
Smoker
non−smoker

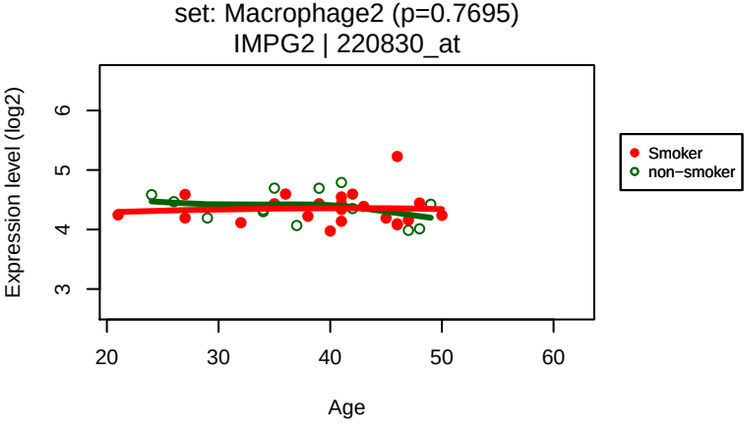

set: SmallAirway2 (p=0.8185)
IMPG2 | 220830_at
Expression level (log2)
Age
Smoker
non−smoker

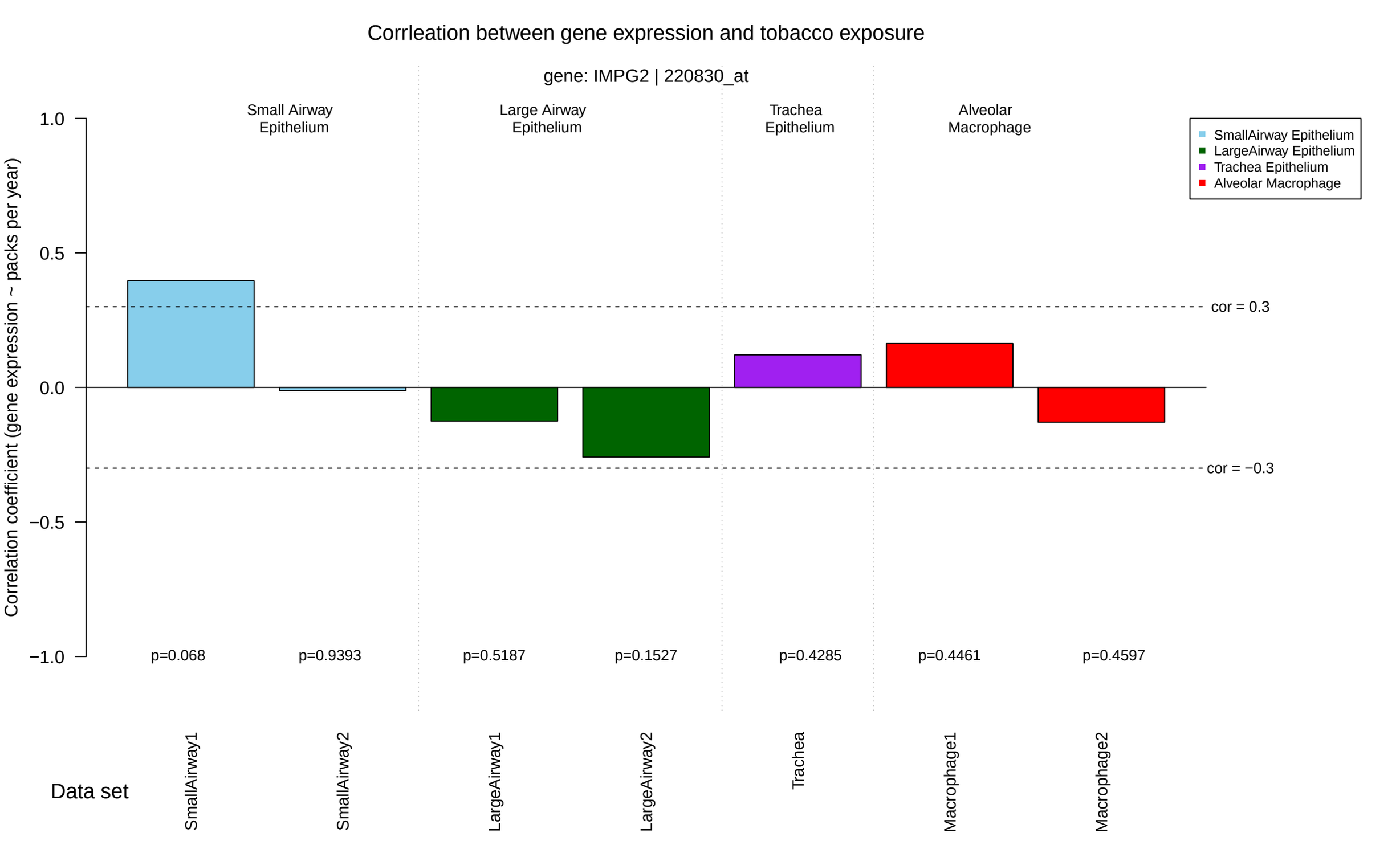

set: LargeAirway1 (p=0.5831)
IMPG2 | 220830_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2192)
IMPG2 | 220830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3067)
IMPG2 | 220830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1633)
IMPG2 | 220830_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.7695)
IMPG2 | 220830_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IMPG2 | 220830_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.068
p=0.9393
p=0.5187
p=0.1527
p=0.4285
p=0.4461
p=0.4597
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2